PHILOSOPHY 1002: Introduction to Critical Thinking
ASSIGNMENTS FOR ALL WEEKS
PART 1

At the end of each week answers to the exercises will be available via LMS.

Week 1: ACE, CARS, ISE or whatever the latest is (the details of these units are to be found on the Unit Outline). Attempt as much as is reasonably possible.
Log in to LMS, (http://www.lms.uwa.edu.au/) and click on the relevant links. These units are compulsory for all newly enrolling Arts, Humanities and Social Sciences students, and advisable for others.) The tasks are not very demanding and it will pay off when you have to write essays.

Week 2: Recognising Argument
I: Govier, pp. 11–13, nos. 3, 5, 11, 19;
Govier, pp. 17–19, nos. 1, 2, 3, 8, 13.
II: In each of the following cases, decide whether or not an argument is offered, giving reasons where possible. If it is, identify the conclusion and any inference indicators.

1. Nobody wants corrupt government. The present government is corrupt, and the opposition is no better. The only way to obtain honest government is to elect the candidates of the Sun-Ripened Warm Tomato Party.
2. Aquinas argued that, since the existence of any object is an empirical question, there can be no *a priori* proof of God's existence.
3. It is an illusion to think that a utopian society could exist without any form of legal regulation. Bertrand Russell argued that even in an anarchist society, there would have to be laws to prevent theft and assault. He believed that even if destitution were eliminated in such a society, some men would want more than others and would, individually or in groups, take it.
4. Either private schools will continue to thrive at a cost to the public system, or the special state funding earmarked for private schools will be discontinued. No party will pay the political price, inevitable when powerful interests are offended, of ending state subsidy of private schooling, so the public system will continue to suffer.
5. This must be a diamond, since it will scratch glass.
6. If sexual orientation is something over which an individual has no control, then discrimination against homosexuals is deplorable, as it is against racial and gender classes, because it holds a person accountable without regard for anything he himself has done. And to hold a person accountable for that over which he has no control is one of the central forms of prejudice. Looking at the actual lived experience of homosexuals in our society, it becomes clear that sexual orientation is not a matter of choice. For coming to have a homosexual identity in our culture does not have the structure of decision-making.
7. In some parts of the USA, parents have objected to high school students' reading *Romeo and Juliet* because it portrays teenage lust, drug use and suicide.
8. Since fundamentalist forms of Christianity have gained more adherents, the government has been under increasing pressure to clarify the doctrine of the separation of church and state.
9. Since the exercise, training and development of our powers of discriminating among works of art are plainly aesthetic activities, the aesthetic properties of a picture plainly include not only those found by looking at it but also those that determine how it is to be looked at.
10. The man who prefers push-pin to poetry may not actually be able to imagine what it would be like to have a developed sensitivity to the nuances of real poetry, and if his intelligence could be raised or his imagination improved, he might be brought to see that he is making a mistake.

<u>Week 3</u>: Argument Structure

I: Take the following arguments from Govier: pp. 32–33, nos. 1, 7, 12, 14, 16. Insert letters and angle brackets, and underline all inference indicators. Then rewrite the arguments in Revised Standard Form, adding (in square brackets) any essential omitted material. (Ignore Govier's own instructions.)

II: Do the same with this argument:

The *Challenger* shuttle explosion could not have been caused by a leaky valve. If the valve had leaked, the instrument panel would have registered it. Since the only other possibility is an expanded, overheated O-ring, that must have triggered the explosion.

<u>Week 4</u>: Argument Evaluation: Support & Acceptability

I: Judge the degree of support in each separate inference in these arguments. Give reasons in each case.

1. Motor vehicle manufacturers say that it is no longer necessary to warm up car motors before driving. Therefore, letting my new car's motor run for a few minutes before driving will not increase its effective life.
2. There were one hundred marbles in this jar, which were thoroughly stirred and mixed before sampling. The first forty marbles that I removed at random were white. Therefore, the remaining sixty are all white too.
3. There were fifty marbles in this jar, which were thoroughly stirred and mixed before sampling. The first forty-nine marbles that I removed at random were all white. Therefore, the remaining marble is white too.
4. The tanks on a ship carrying liquefied natural gas must have double walls and insulation more than a metre thick to reduce the rate of vapourisation. Accordingly, a collision that might lead to a loss of cargo would have to be considerably more severe than it would if the ship were built like a petroleum tanker, where the tank is the hull of the ship itself. So accidents like ramming and grounding are highly unlikely to cause the release of cargo from a ship carrying liquefied natural gas.

II: Answer the following questions, and explain your answers.

1. Can a valid argument lead to a false conclusion?
2. From the fact that an argument has all its premises and conclusions true, can we tell anything about its validity?
3. If a true conclusion follows from the premisses of an argument, must all those premisses be true as well?
4. If a situation in which the premisses of an argument are true and the main conclusion false cannot be described without self-contradiction, is the degree of support which those premisses give to that conclusion complete or nil?
5. If a situation in which the premisses of an argument are true and the conclusion false is quite probable, do those premisses give weak or strong support to that conclusion?

III: In each of the following cases, determine the degree of support which the premisses give to the conclusions; then decide whether the support is formal or non-formal. If it is formal, then express the form using schematic letters.

1. Some children are hyperactive, because all hyperactive people are difficult to handle and some children are difficult to handle.
2. Sceptics are not reluctant to admit their own ignorance. They know that full knowledge is impossible, and everyone who knows that full knowledge is impossible willingly admits his own ignorance.
3. Only those who have at least one blue-eyed parent are blue-eyed. Some Irish are green-eyed. So some Irish do not have at least one blue-eyed parent.
4. He is either stupid or wicked, and he's certainly not stupid.
5. If Marcus comes, the party will be a disaster. And if the party's a disaster, we'll lose the contract. But the fortune teller told me that we're going to make a lot of money out of this deal. So Marcus won't come to the party.

IV: Govier, pp. 128–30: Part A, nos. 1, 2 (substituting 'Determine whether or not it is an *a priori* truth and explain the basis for your answer' for Govier's own instruction); Part B, nos. 3, 5, 8.
V: Govier, pp. 135–39: Part A, no. 6; Part B, no. 6; Part C, nos. 4, 9.

<u>Week 5</u>: Proof; Conditionals
I: Govier, pp. 100–3, no. 2.
II: Put the following arguments into revised standard form and evaluate them:
a. "Marijuana should not be illegal, because it is less harmful than alcohol and alcohol is legal."
b. "If Hitler had died at the age of ten, Nazism would not have developed in Germany in the way it did. If Woodrow Wilson had never existed, there would have been no League of Nations. And without Mao Zedong, who knows what would have come of the revolutionary movement in China during the 1930s and 1940s? From these facts, it follows that particular individuals determine the course of world history."
III: Answer the following questions, giving reasons for your answers.
1. Does a sound argument always prove its conclusion?
2. Is it true that an argument is cogent iff all its premisses are acceptable and they give some degree of support to the conclusion?
3. Is a cogent argument a proof of its conclusion?
4. Is a sound argument always deductively valid?
5. Do unsound arguments have false conclusions?
6. Must an argument with a false conclusion be (a) unsound, (b) not cogent?

IV: Conditional 1: 'If a substance is fluorine, then it has an atomic number of 9.'
1. Rewrite conditional 1 in its logically equivalent 'only if' form.
2. Rewrite conditional 1 in its contraposed (transposed) form.
3. Convert conditional 1 into an explicit statement of a sufficient condition.
4. Convert conditional 1 into an explicit statement of a necessary condition.
5. Rewrite conditional 1 in the form of a 'Not ... , unless ... .' sentence.
6. Write out the converse of conditional 1.
7. Write out an argument in revised standard form which has conditional 1 as sole premiss and the converse form as conclusion. Is the argument valid? Give reasons for your answer.
8. Write out an argument in revised standard form which has conditional 1 as sole premiss and the transposed form as conclusion. Is the argument valid? Give reasons for your answer.
9. Take a valid argument from 7 or 8, and write out the necessarily true conditional which results from the application of the technique of conditionalisation to the argument.
10. Write out the transposed version of the conditional in 9.
11. Write out the biconditional formed by combining conditional 1 above and its converse.

V: What is the fallacy in the following piece of reasoning?
'In the early fourteenth century was issued a royal charter in which it was formally . . . decreed that only members of a craft might be admitted to the freedom of the city. Only citizens, in other words, could belong to a trade guild.'
[Peter Ackroyd, *London: The Biography* (Chatto and Windus, 2000), p. 90]

Hints: (1) Assume that membership of a craft and belonging to a trade guild are the same thing, and that becoming a citizen and being admitted to the freedom of the city are the same thing. (2) Rewrite the 'only' statements as conditionals in the way suggested in *Lecture Notes 5A.*

<u>Week 6</u>: Argument Construction
This assignment gives you some practice in the construction of arguments.
<u>Part A</u>
Listed below are three groups of sentences. One member of each group recommends some course of action. For each group, make this sentence your main conclusion, and then, using the letters provided, arrange the others in the most natural way possible to form an argument for that conclusion. Express your arguments in revised standard form.

I: a. It is important to have a trained army for purposes of defence.
b. The Federal Government should reintroduce conscription.
c. Having an entirely volunteer defence force is unfair.
d. Having an entirely volunteer defence force effectively conscripts the already underprivileged who have nowhere else to go.
e. Conscription would reduce unemployment.
f. Conscription would provide many young people with skills and experience that will help them find jobs in civilian life.

II: (In this case, concentrate your efforts on laying the argument out and do not spend a lot of time on trying to provide bracketed derivations, as they are very difficult. This does not mean that you are to spend no time at all on the matter.)
a. If God does not exist, you will lose nothing.
b. If God does not exist, you will lose very little.
c. Suppose you believe in God's existence and live a Christian life.
d. You should believe in God's existence and live a Christian life.
e. If God exists, you will enjoy eternal bliss.
f. If God exists, you will suffer eternal damnation.
g. Either God exists or God does not exist.
h. Suppose you don't believe in God's existence and don't live a Christian life.

III: a. It is unscientific to accept creationism.
b. Evolutionary theory remains the best available explanation for the origin of species.
c. If we teach creationism, we should also teach all the other versions of creation contained in religion and myth.
d. The only reason for introducing creationism is to illustrate accounts that rely on something other than scientific reasoning.
e. Portraying the adoption of a scientific theory as merely a matter of individual choice unnecessarily confuses a school-age child.
f. Creationism should not be taught as an alternative to evolutionary theory.
g. Positive arguments in favour of creationism ultimately rest on reference to the Book of Genesis rather than on empirical evidence.

Part B

Construct an argument by *reductio ad absurdum* against the following claim:
Every religion is true.

ASSIGNMENTS FOR PART 2

You should do the exercises in the relevant chapters in the text-book, especially the ones for which answers are provided. The following examples are mainly ones from Govier, those for which she does not provide answers.

Week 7: Definitions & Clarity
(a) Govier, Ch. 3: Ex. 3, pp. 79–80: Q. 3: f, i, j.
(b) Check the following passages to see whether they contain examples of vagueness, ambiguity, euphemisms or emotionally charged language. Is the reasoning affected by any of these things? [Do not assume that each passage contains some of these examples.]

1. *"Art is energy. It is a privileged communication that passes between something and the spirit if a human."*
2. *"Physicians in general do not seem to commit suicide at area significantly different from that of their non-medical peers, although perhaps of their knowledge of drugs and access to them, their methods of choice are characteristically non-violent: doctors poison themselves more than twice as often as the lay public, and shoot themselves less often, Psychiatrists, however, show a markedly greater tendency to commit suicide than the population at large or their medical peers."*

3. *"The probation officer told her client that if he was not able to keep appointments it would be necessary to consider a reinstatement of his previous situation with reference to penal institutions. The client asked whether he would have to go back to jail."*
4. *"Children who are unable to move on to the next grade do not fail. They are retained."*

Week 8: Fallacies of Clarity, and Relevance

(a) For each of the following passages, (i) determine the conclusion (if any) and then (ii) indicate any premises that you consider irrelevant to the support of that conclusion. Then (iii) state why you think the premise is irrelevant.

1. "Children are unique and sensitive creatures. They are very imaginative, and they are different from adults. Therefore, every child has an absolute right to state-supported education."

2. [Background: the following was written in response to a proposal to license cats.] "Cats are free spirits, the last really independent creatures around. You can no more license cats than you can license the wind. Dogs may submit to bureaucracy. Cats won't. The same spirit tends to rub off on cat owners. They have enough trouble being pushed around by their cats without being asked to submit to man-made laws. Besides, there's an economic factor. They've never had to buy licenses, so why start? No, it just won't work." [Quoted by Ralph Johnson in "Charity Begins at Home," Informal Logic Newsletter, [June 1981], pp. 4-9]

(b) Govier, Ch. 6: pp. 167–70, Ex. 2, Q. 4. [Check the answer in the text: what do you think?]

Week 9: Test #1 Self-Assessment

Assess your performance on the first take-home test. Be sure that you understand the correct answers and take careful note of your errors. Devote additional time to the study of material you have not mastered adequately. Write down specific questions to be taken to your workshop group for discussion.

Week 10: Analogies

(a) Govier, Ch. 11: pp. 329–33: Ex. 1, Part A: Q. 2; Part B: Q. 12, Q. 15;
(b) Govier, Ch. 11: pp. 335–38: Ex. 2, Q. 2;
(c) Govier, Ch. 11: pp. 345–49: Ex. 3, Q. 15.
(d) For the following passages, first identify whether it contains an argument by analogy. If it does, identify the primary subject and the analogue, mention key relevant similarities and difference between the primary subject and the analogue, and comment on the merits of the argument. If a passage contains no analogy, or contains an analogy that is not used for the purpose of argument, say so and explain the basis for your interpretation.

1. "Handgun control doesn't necessarily mean taking the guns away from everybody. It can mean simply to license these weapons, making it unlawful to own one without proper registration. After all, what's the big deal? You need a license to get married. You need a license for your dog. You need one for your vehicle and your business. You need permits for nearly everything. Nobody seems to suffer too much. Drivers must meet certain standards in order to obtain a permit to drive. As a result, thousands of lives are saved every year. So why not similarly license handguns? It'll cost a little, be a little inconvenient, and maybe it'll save a few lives. It really is the least we can do." [Letter to the editor, Los Angeles Times, January 23 1981, cited in the Informal Logic Newsletter, Examples Supplement, November 1981.]

2. [Background: The following passage is taken from a self-help book with a spiritual orientation.] "Anytime you have a negative feeling toward anyone, you're having an illusion. There's something seriously wrong with you. You're not seeing reality.

Something inside of you has to change. But what do w generally do when we have a negative feeling? 'He's to blame, she is to blame. She's got to change.' No! The world is all right. The one who need to change is you.' [Anthony de Mello, Awareness: The Perils and Opportunities of Reality (New York: Doubleday Image Book, 1990), p. 51)

Week 11: Inductive Arguments

(a) Govier, Ch. 9: pp. 265–68: Ex. 1, Part A: Q 4; Part C: Q. 7 & Q. 9.

(b) Govier, Ch. 10: pp. 310–14: Ex. 3: Q. 1, Q. 5 & Q. 9.

Week 12: Conductive Arguments

(a) Govier, Ch. 12: pp. 366–70: Q.9, Q. 12 & Q. 16.

(b) Do the second take-home test.

Test #2 Self-Assessment

Assess your performance on the second take-home test. Be sure that you understand the correct answers and take careful note of your errors. Devote additional time to the study of material you have not mastered adequately. Write down specific questions to be taken to your tutor for clarification.